

LoRC: Low-rank Quantization Correction

1 Introduction

We consider the problem of compressing a pretrained language model f_θ while preserving its performance on a target domain $\mathcal{D}_{\text{target}}$ at the expense of out-of-domain generality. Our approach – **Low-rank Quantization Correction** – identifies low-dimensional subspaces where activation statistics differ between two domains and uses them to correct quantization error.

1.1 Covariance estimation

For a linear layer with weight $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, let $x \in \mathbb{R}^{d_{\text{in}}}$ and $y \in \mathbb{R}^{d_{\text{out}}}$ be its pre- and post-activation respectively. For each domain $\mathcal{D}$, we collect the second-moment statistics

$$C_{\text{pre}}^{\mathcal{D}} = \mathbb{E}[xx^\top] - \mathbb{E}[x]\mathbb{E}[x]^\top \in \mathbb{R}^{d_{\text{in}} \times d_{\text{in}}}, \quad (1)$$

$$C_{\text{post}}^{\mathcal{D}} = \mathbb{E}[yy^\top] - \mathbb{E}[y]\mathbb{E}[y]^\top \in \mathbb{R}^{d_{\text{out}} \times d_{\text{out}}}, \quad (2)$$

estimated by accumulating $\sum xx^\top$ and $\sum x$ over a stream of domain-specific prompts.

1.2 Domain subspace extraction

Given two domains $\mathcal{D}_1$ and $\mathcal{D}_2$ (e.g., Lean4 proofs and Wikipedia text), form the difference

$$\Delta C = \alpha C^{\mathcal{D}_1} - \beta C^{\mathcal{D}_2}, \quad \alpha, \beta \geq 0, \quad (3)$$

symmetrize as $(\Delta C + \Delta C^\top)/2$, and compute the eigendecomposition

$$\Delta C = Q\Lambda Q^\top. \quad (4)$$

The eigenvectors with the largest eigenvalues correspond to directions where $\mathcal{D}_1$ activates more than $\mathcal{D}_2$ (the $\mathcal{D}_1$ -dominant subspace); those with the most negative eigenvalues correspond to the $\mathcal{D}_2$ -dominant subspace. We keep $K \ll d$ eigenvectors in each direction, forming the column-orthonormal matrices $V_1, V_2 \in \mathbb{R}^{d \times K}$.

1.3 Correction factors

Quantize the weight matrix to NF4 and compute the quantization error

$$E = W - Q^{-1}(Q(W)). \quad (5)$$

Project E onto the $\mathcal{D}_1$ subspace (the $\mathcal{D}_2$ case is symmetric). For a weight applied on the input side (e.g. up-projection), the projection is

$$E_{\text{proj}} = E (V_1 V_1^\top), \quad (6)$$

and for a weight applied on the output side (e.g. down-projection)

$$E_{\text{proj}} = (V_1 V_1^\top) E. \quad (7)$$

Factorize $E_{\text{proj}} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ via the singular value decomposition

$$U \Sigma V^\top = \text{SVD}(E_{\text{proj}}), \quad (8)$$

$$U_k \Sigma_k V_k^\top \approx E_{\text{proj}}, \quad (9)$$

truncated to rank $k = \min(K, d_{\text{out}}, d_{\text{in}})$. Define the correction factors

$$V_{\text{act}} = V_k \Sigma_k^{1/2} \in \mathbb{R}^{d_{\text{in}} \times k}, \quad (10)$$

$$U_{\text{write}} = U_k \Sigma_k^{1/2} \in \mathbb{R}^{d_{\text{out}} \times k}, \quad (11)$$

both stored in bfloat16.

1.4 Hybrid forward pass

At inference, the base weight remains in NF4. The forward pass becomes

$$z = x W_{\text{base}}^\top + (x V_{\text{act}}) U_{\text{write}}^\top, \quad (12)$$

where the first term is computed via on-the-fly dequantization of the 4-bit weights and the second term supplies the low-rank BF16 correction for the active domain.

1.5 Disjunction score

To measure whether the identified subspaces are genuinely domain-specific, we project them out of the original weight matrix. For weights applied on the input side (e.g. up-projection):

$$W' = W - W(VV^\top), \quad (13)$$

and for weights applied on the output side (e.g. down-projection):

$$W' = W - (VV^\top)W. \quad (14)$$

We then measure the perplexity change ΔPPL on each domain. The disjunction score

$$D = \frac{\max(\Delta\text{PPL}_{\text{own}}, \varepsilon)}{\max(\Delta\text{PPL}_{\text{other}}, \varepsilon)} \quad (15)$$

quantifies domain specificity: $D \gg 1$ indicates genuine separation, while $D \approx 1$ indicates polysemantic (entangled) directions.

1.6 Possible improvements

The current implementation incurs $\mathcal{O}(d^2)$ memory for covariance storage and $\mathcal{O}(d^3)$ for the eigendecompositions, which limits scalability to larger models. Several conceptually motivated optimizations exist.

Streaming PCA. Instead of accumulating the full outer-product matrix $\sum xx^\top$, maintain a subspace estimate $V \in \mathbb{R}^{d \times K}$ via online updates (Oja’s rule):

$$V \leftarrow V + \eta(xx^\top V), \quad V \leftarrow \text{QR}(V), \quad (16)$$

where x is a sampled activation. Storage drops from $\mathcal{O}(d^2)$ to $\mathcal{O}(dK)$, and the two domains can be processed concurrently using domain labels.

Randomized eigendecomposition. Only the K extremal eigenvalues of ΔC are needed. Replace the full $\text{eigh}(\Delta C)$ with a Krylov subspace method:

$$\Omega \sim \mathcal{N}(0, 1)^{d \times (K+p)}, \quad Y = \Delta C \Omega, \quad Q = \text{QR}(Y), \quad (17)$$

$$T = Q^\top \Delta C Q \in \mathbb{R}^{(K+p) \times (K+p)}, \quad \tilde{\Lambda}, \tilde{U} = \text{eigh}(T), \quad V_{\text{ext}} = Q\tilde{U}, \quad (18)$$

with oversampling $p \approx 5$ for numerical stability. The cost is $\mathcal{O}(Kd^2)$ rather than $\mathcal{O}(d^3)$.

Avoiding the explicit projection matrix. The projected error $E_{\text{proj}} = E(VV^\top)$ is never needed explicitly. Factorize $A = EV \in \mathbb{R}^{d_{\text{out}} \times K}$ directly:

$$U_A \Sigma_A R_A^\top = \text{SVD}(A), \quad (19)$$

$$V_{\text{act}} = (V R_A) \Sigma_A^{1/2}, \quad U_{\text{write}} = U_A \Sigma_A^{1/2}. \quad (20)$$

This produces identical correction factors without constructing the $d_{\text{in}} \times d_{\text{in}}$ projection matrix and without a full-size SVD. Total cost reduces from $\mathcal{O}(d^3 + d^2 K)$ to $\mathcal{O}(d^2 K)$.

Analytic causal masking. The singular values Σ_A of $A = EV$ are the per-component contributions to the projected quantization error. Gradient-based mask training can be replaced by a simple threshold:

$$\text{keep}_i = \mathbf{1}(\sigma_i > \tau), \quad \tau = \text{quantile}(\Sigma, 1 - \text{keep_pct}), \quad (21)$$

eliminating the data-dependent optimization loop entirely.

Extension to more than two domains. For $D > 2$ domains, replace the pairwise difference ΔC with multi-class Fisher discriminant analysis:

$$W^* = \arg \max_{W^\top W = I} \frac{\text{tr}(W^\top C_{\text{between}} W)}{\text{tr}(W^\top C_{\text{within}} W)}, \quad (22)$$

where C_{between} is the between-class scatter matrix and C_{within} is the pooled within-class covariance. The optimal W is given by the generalized eigenvectors of $C_{\text{between}} v = \lambda C_{\text{within}} v$, yielding a single subspace that maximally separates all D domains.

Cross-layer subspace coupling. Per-module subspaces are estimated independently. Constraining them to share a global basis $V_{\text{gl}} \in \mathbb{R}^{d \times K}$ with per-module adapters M_ℓ such that $V_\ell = V_{\text{gl}} M_\ell$ reduces overfitting and total parameter count, especially in deeper models where domain-specific patterns recur across layers.